

## **Fluid Flow in Pipes**

Pre-Lab Plan Unit Operations Lab #6

November 12<sup>th</sup>, 2025

Group 3, Tuesday, PM section

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## 1. Experimental Objective (5 pts)

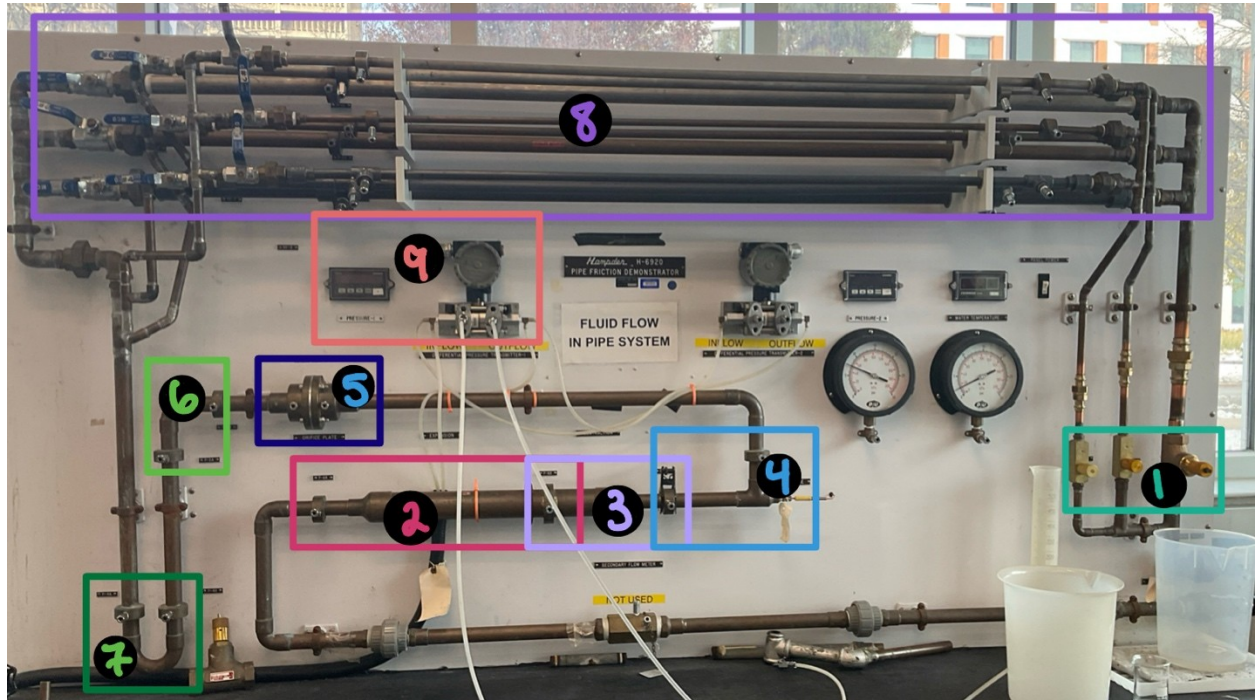
In this lab, experimental coefficients including friction factor, loss coefficient, and flow coefficient will be obtained for a set of fluid flow experiments that demonstrate typical flow patterns through different materials, bends, and fittings. These experimental values will be compared to known empirical correlations. Pressure drop and water flowrate will be used to complete these calculations and comparisons.

## 2. Experimental

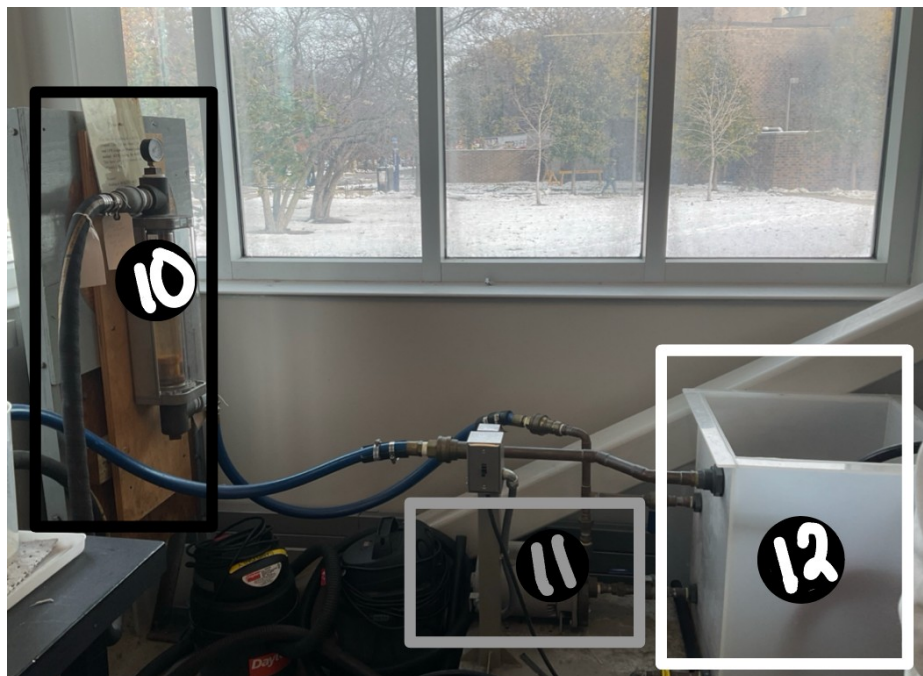
Within this lab, the different experiments utilize different sections of pipe within the H-6920 Pipe Friction Demonstrator apparatus (**Figure 1**). To begin the experiments, the reservoir tank (**#12**) attached to the system should be filled with water above the pump outlet and the pump (**#11**) should be turned on. Once water has started flowing through the system, the valves in the apparatus (**#1 and #8**) can be opened/closed to direct flow through different pipes. Avoid closing all valves in order to prevent a build up of pressure due to a lack of an outlet in the system.

The straight pipes of different sizes and materials (**#8**) are utilized to test flow through different pipe diameters and materials of construction. The different bends and fittings in the lower part of the apparatus can be used to complete tests that correspond to the labels above each fitting. To measure pressure drop, the free end of the tubes labelled 'inlet' and 'outlet' on the instrumentation panel (**#9**) should be attached to the inlet and outlet of the length of pipe being tested. The pressure drop (in psid) can be read on the left-most display on the instrumentation panel, and the inlet/outlet pressures can be read on the right-hand side using either the digital displays or the analog pressure gauges. Water flowrate through the system can be measured using the rotameter (**#10**) on the right-hand side of the apparatus (although calibration is needed), and controlled using the three valves on the vertical portion of the pipe circuit (**#1**).

## 2.1 Apparatus (5 pts)



**Figure 1:** Main components of the apparatus



**Figure 2:** Additional apparatus parts.

**Table 1:** Relates the color-coded labels from **Figure 1** and **Figure 2** to what part of the apparatus they are.

| Number/Color     | Apparatus Part                                       |
|------------------|------------------------------------------------------|
| 1 – Teal         | Water flow control knobs                             |
| 2 – Pink         | Expansion                                            |
| 3 – Light Purple | Contraction                                          |
| 4 – Light Blue   | Square corner                                        |
| 5 – Dark Blue    | Orifice Plate                                        |
| 6 – Light Green  | Elbow                                                |
| 7 – Dark Green   | U - tube                                             |
| 8 – Dark Purple  | A variety of pipes varying in diameter and material. |
| 9 – Orange       | Digital pressure drop gauge                          |
| 10 – Black       | Rotameter                                            |
| 11 – Grey        | Pump                                                 |
| 12 - White       | Water reservoir                                      |

## 2.2 Materials and Supplies (5 pts)

**Table 2:** A list of the materials/supplies used for lab.

| Material/Supplies | Purpose/Specifications                                                                                                                                    |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| City Water        | This is the water that will flow through the system                                                                                                       |
| Tubes with clips  | To be used at different inlet/outlets in conjunction with apparatus part nine in order to read the pressure drop across the aspect.                       |
| Beaker            | Used to collect water flowing through the system at different flow rates in order to calculate a correction factor for the rotameter (apparatus part ten) |

## 2.3 Experimental Plan (10 pts)

- [Prelab] Calibrate the rotameter by filling the reservoir with water, turning on the pump, and collecting 3 volumes over 3 timed intervals for 3 different pressures.
- On lab day, fill the reservoir with city water, ensuring the drain valve is closed.
- Turn on the pump and instrumentation and allow 3-5 minutes for the system to run and warm up.
- Record pressure and water temperature using the panel thermocouple.
- Zero the differential pressure transmitters (DPTs). If needed, open/close their side valves to flush and re-zero.
- Confirm all pressure taps are capped and verify there are no leaks at quick disconnects.
- Decide the run sequence and set up a shared data sheet.

- Connect the DPT across the orifice meter pressure taps (upstream to high, downstream to low).
- For 8-10 flow rates spanning as wide a Reynolds number range as achievable, set the flow and confirm steady  $\Delta P$  across the orifice.
- Log  $\Delta P_{\text{orifice}}$ , rotameter reading, temperature, and also measure collected-volume  $Q$  on 3 representative points to spot check the rotameter calibration.
- Repeat two the intermediate points once to estimate short term repeatability.
- Between points, briefly close the valve, move/flush lines as needed, and re-open to steady state.
- Select a straight test section, connect DPT across its taps
- For each pipe, sweep  $\sim 10$  flow settings from very low to high to span laminar, transition, and turbulent, when possible with available pump head.
- At each point, wait for steady state, record  $\Delta P_{\text{pipe}}$ , rotameter reading, and temperature.
- Repeat a mid-range point and the first point at the end (reverse sweep) to check hysteresis/drift.
- Switch to the next pipe, close the valve, disconnect quick couplers, reconnect at the next tap pair, flush DPT, then resume.
- Connect DPT across each fittings taps for the bend, elbow, expansion/contraction, and U-tube.
- For 8-10 flows per fitting, stabilize flow, record  $\Delta P_{\text{fit}}$ , rotameter reading, and temperature.
- Note the valve positions to keep upstream approach conditions consistent between fittings.
- Repeat one mid-range point to assess repeatability for each fitting.
- Return the valves to safe positions, release line pressure, power off instruments and pump.
- Drain water by opening the drain valve and allow water to filter out into the floor drain.

The independent variable in all runs is the volumetric flow rate, spanning the widest feasible range to achieve  $Re \approx 5 \times 10^2$  to  $\geq 5 \times 10^4$  depending on diameter. Dependent variables include the measured pressure drops across the orifice, straight pipes, and fittings, from which  $C_f$ ,  $f$ , and  $K$  are computed. Water temperature is logged each point to evaluate  $\rho$  and  $\mu$ . Flow is measured by collection and time during calibration and then by the calibrated rotameter. Each distinct condition includes  $\geq 1$  replicate, with a 95% confidence interval reported for calibration and propagated to derived coefficients. Goodness of fit will be assessed using  $R^2$ , RMSE, residual diagnostics, and lack of fit tests for the calibration. Final data will be correlated to Bernoulli + loss models for orifice and fittings and to Darcy-Weisbach (Moody/Colebrook) for straight pipe friction, using the manual's geometry and roughness values.

## **Project Safety Evaluation (10 pts)**

Although there are no toxic chemicals used in this experiment, there are several safety hazards to consider when running the experiment. The working fluid is water recirculated through a bench-mounted Hampden H-6920 panel that includes multiple straight pipes and fittings instrumented with pressure taps, a rotameter, and differential-pressure transmitters (DPTs). Key process conditions are ambient-temperature water with pressure differentials read directly in  $\Delta$ psi across individual components and flow indicated in LPM water after calibration. Primary hazards include hydraulic energy/over pressure localized across restrictions and along straight test sections where taps are connected. These are the places where sudden valve movements or mis-plumbed lines could create unexpected  $\Delta$ P spikes and leaks.

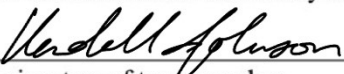
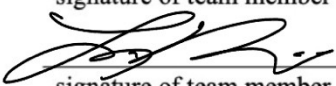
Splash and slip hazards from hose/tap disconnections during changeovers and electrical hazards from operating the pump and digital instrumentation in a wet environment. The risk controls are embedded in both the hardware and the procedures. Flow will always be isolated before moving the quick-disconnect tubing, then taps will be reconnected with upstream to the DPT high side and downstream to the low side. The section valve should be opened slowly to re-establish flow, and  $\Delta$ P will be read only after the signal is ready. This sequence minimizes pressure shocks and exposure to leaks. Before each measurement, DPTs will be verified at readings of  $\sim 0.00$ - $0.002$   $\Delta$ psi at zero and flush/re-zero using the two side black valves if needed. This reduces instrument error and prevents water ingress or cross-contamination in the transmitters. Flow will be set with the control valve and monitored on the rotameter.

Temperature will be logged at the panel thermocouple so fluid property changes are recognized. During all switchovers, the panel's stepwise method will be strictly followed: close the section valve, pull the quick-disconnects from the sensing ports, connect to the next ports, open the valve to start flow, flush and close the DPT's black valves, then wait for steady readings. Both hydraulic and splash risks are controlled at the source. Good lab practice supplements these controls: keep cords and meters off wet surfaces, use towel trays/absorbents beneath disconnects, wear safety glasses and close-toed shoes, maintain drip loops on power leads, and never bypass guards or operate with missing tap caps.

Process parameter envelopes are maintained well within instrument and panel limits by watching  $\Delta$ psi and flow while approaching target points gradually. If chatter, cavitation sounds, or unstable  $\Delta$ P appear, re-establishing steady conditions will be implemented. Collectively, these controls will reduce the likelihood and consequences of the credible hazards in this water-flow experiment across the orifice, straight-pipe, and fittings.

|                                     |                     |                                                                    |
|-------------------------------------|---------------------|--------------------------------------------------------------------|
|                                     | <i>Yes/No</i>       |                                                                    |
| Potential electrical hazards?       | Y                   | If yes, identify conditions: Electric comp. near water             |
| Potential mechanical hazards?       | Y                   | If yes, identify conditions: Pump                                  |
| Potential pressure hazards?         | Y                   | If yes, identify conditions: If pump is dead headed to system      |
| Potential temperature hazards?      | N                   | If yes, identify conditions:                                       |
| Hazardous/toxic chemicals involved? | N                   | If yes, hazards involved:                                          |
| Gloves required?                    | N                   | If yes, specify type:                                              |
| MSDS reviewed by all team members?  | N                   | List MSDS sheets reviewed:                                         |
| <i>Process Parameters</i>           | <i>Max Expected</i> | <i>List location and possible hazards and precautions</i>          |
| Temperature (°C)                    | RT                  |                                                                    |
| Pressure (psia)                     | 50                  | Based on pressure read on rotameter when finding correction factor |
| <i>Safety Controls</i>              | <i>yes/no</i>       | <i>If yes, Provide description</i>                                 |
|                                     | Y                   | Pump shut off, various valves                                      |

I have read relevant background material for the Unit Operations Laboratory entitled: "Fluid Flow in Pipes" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

|                                                                                      |          |
|--------------------------------------------------------------------------------------|----------|
|  | 11/12/25 |
| signature of team member                                                             | date     |
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**Figure 3:** Safety evaluation chart.